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Predicting halo triaxility in cosmological simulations using a convolutional neural network

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ABSTRACT

Aims. This aim of this work is to predict dark matter halo geometry directly from the initial linear density field in N-body simulations using a 3D convolutional neural network (CNN).
Methods. Training, validation and test datasets were constructed from cosmological N-body simulations generated with `DISCO-DJ` (Hahn et al. 2024; List & Hahn 2024), with halos and their axis ratios (b/a and c/a) identified using the `ROCKSTAR` halo finder (Behroozi et al. 2012). A 3D U-Net CNN was trained on normalized, augmented sub-volumes using a Mean Squared Error (MSE) loss function. The predicted axis ratios were compared against ground-truth values using voxel-wise metrics and visual comparisons.
Results. While the model successfully identified collapsed regions on large scales, it failed to predict halo axis ratios accurately. Instead, predictions collapsed to the mean near halo centers with artificial, decreasing gradients toward halo boundaries, resulting in negative coefficient of determination values; $R^2 = -2.45$ for both axis ratios. Single-sample overfitting diagnostics confirmed that the network struggles to generalize geometric features across independent simulations. Furthermore, masking background voxels to isolate halo shapes caused predicted values to scatter tightly around the dataset mean, resulting in $R^2 \approx 0$.
Conclusions. The CNN under this configuration cannot extract physically meaningful halo axis ratios from initial density fields. Future improvements likely require transitioning to alternative architectures, using physics-informed loss functions.

Key words. halo triaxiality, large-scale structure, neural networks

1. Introduction

N-body simulations have demonstrated that dark matter halos formed through gravitational collapse are non-spherical (Jing & Suto 2002), showing shapes that depend on halo mass (Bonamigo et al. 2015; Jing & Suto 2002), assembly history and cosmic environment (Macciò et al. 2007; Lee et al. 2017). Determining halo geometries accurately is essential because halo shapes directly influence key physical properties and observations, such as subhalo orbital dynamics, spatial anisotropies in satellite galaxy distributions (Zentner et al. 2005), two-point correlation functions (van Daalen et al. 2012) weak gravitational lensing signals (Corless & King 2007) and more. Therefore, mapping the origin and distribution of halo shapes provides crucial insights into the non-linear growth of cosmic structure and is vital for accurate cosmological parameter inference.

While N-body simulations provide the ground truth for extracting halo geometries, running full numerical simulations and halo-finding algorithms is computationally expensive. To overcome this computational bottleneck, the application of machine learning architectures to cosmological N-body simulations has expanded significantly in recent years. Some examples of promising results include predicting non-linear structure evolution (Jamieson et al. 2023), cosmic web features (Lahiry et al. 2025), line-of-sight velocities (Chen et al. 2024), halo assembly bias (Lucie-Smith et al. 2023), and halo mass distributions (Buisman et al. 2025; Lucie-Smith et al. 2024). This motivates the use of neural networks to predict halo geometry.

This work is part of a larger project predicting halo properties from the initial density field. While Jonas & Einramhof (2026) investigates the prediction of structural collapse and the halo mass function, this analysis focuses on halo axis ratios. Both works use a similar U-Net convolutional neural network to make voxel-level predictions for each particle. This was implemented using the `Pytorch` library (Paszke et al. 2019). `Disco-DJ` (Hahn et al. 2024; List et al. 2026) was used to run N-body simulations and `ROCKSTAR` (Behroozi et al. 2012) was used to create training and evaluation data. The model's predictions were then compared against the ground truth to evaluate its performance. This model was evaluated on binary collapse classification and compared to other works such as López-Cano et al. (2024) and Buisman et al. (2025), as well as voxel-wise scatter plots.

This work is structured as follows: section 2 introduces the label generation, data preprocessing, convolutional neural architecture, and the evaluation metrics. section 3 shows the results for binary classification, unmasked and masked shape predictions, which will subsequently be discussed in section 4. Lastly, section 5 will provide a short conclusion and outlook.

2. Methods

2.1. Generating data

This section outlines the process of generating the data used by the CNN to predict the halo triaxility.

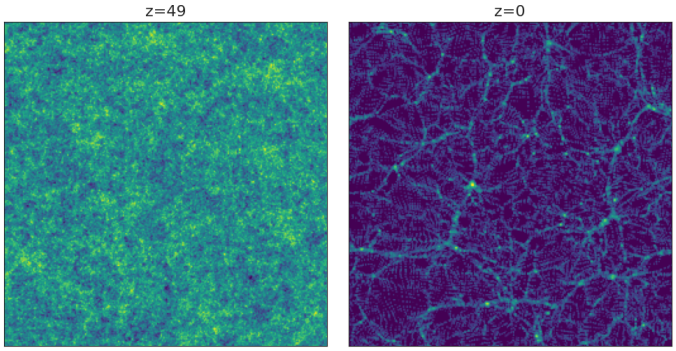


Fig. 1. Initial field at $z = 49$ (left) and evolved field at $z = 0$ (right) for an example simulation generated using `DISCO-DJ`.

2.1.1. Generating simulations using `DISCO-DJ`

The differentiable software package for large-scale structure cosmology `DISCO-DJ` (DIfferentiable Simulations for COsmology - Done with JAX) (List et al. 2026; Hahn et al. 2024) was used to generate 110 N-body simulations. The simulation set-up closely follows the work of Buisman et al. (2025); they were set up with a box size of $L = 100$ Mpc$/h$, $N = 256^3$ particles and a PM grid with $N_g = 1024^3$ cells. Furthermore, the fiducial flat ΛCDM cosmology was used; $\Omega_c = 0.259605, \Omega_b = 0.0488911, h = 0.67742, n_s = 0.96822, \sigma_8 = 0.808992$. The simulation was initialized using second-order LPT at $a = 0.02$ ($z = 49$) and evolved to $a = 1.0$ ($z = 0$) using the `BULLFROG` N-body integrator (Rampf et al. 2025) in 100 time steps. Figure 1 shows the initial and evolved field for an example simulation.

The initial field of the simulations was saved to serve as input for the CNN. Moreover, the particle positions and velocities at $z = 0$ and their IDs were stored in an `.ascii` file to be further processed using the `ROCKSTAR` halo finder (Behroozi et al. 2012).

2.1.2. Identifying halos using `ROCKSTAR`

`ROCKSTAR` (Robust Overdensity Calculation using K-Space Topologically Adaptive Refinement) is a halo finder based on adaptive hierarchical refinement of friends-of-friends groups in six phase-space dimensions and one time dimension (Behroozi et al. 2012). It was used to identify halos in the simulation box and assign particles to them. For this, two `ROCKSTAR` outputs were necessary: the halo catalog (`.ascii`) file, which contains halo properties and the particle table (`.particles`), which contains particle positions, velocities and IDs, as well as halo IDs. From the halo catalog, the position in three dimensions [Mpc$/h$], the virial mass [$M_\odot/h$], the virial radius [kpc$/h$], the axis ratios b/a and c/a and the parent IDs were extracted for each halo. From the particle table, the particle IDs, assigned halo IDs, internal halo IDs and external halo IDs were extracted for each particle. Figure 2 visualizes a 10 Mpc/h thick density slice at z=0 with the identified rockstar halos marked on top for an example simulation.

2.1.3. Generating labels

The shape of a halo can be quantified through its triaxiality value

$$T = \frac{1 - \frac{b^2}{a^2}}{1 - \frac{c^2}{a^2}} \tag{1}$$

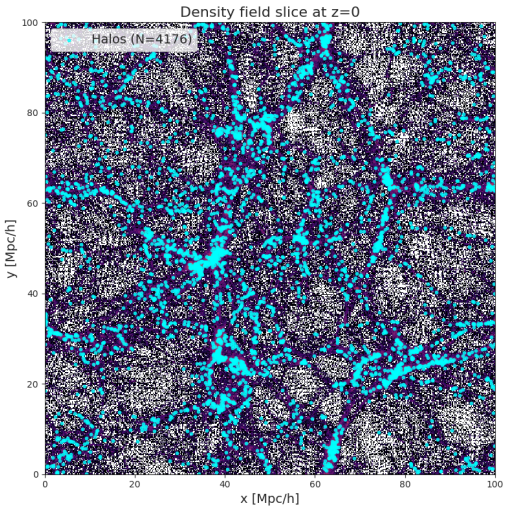


Fig. 2. Density slice (20-30 Mpc/h) of an example simulation at z=0. The identified `ROCKSTAR` halos are marked in cyan.

(Giocoli et al. 2026). To compute this, each particle in the simulation volume must be assigned its host halos axis ratios b/a and c/a. The distribution of the values of both axis ratios and triaxiality of all 110 simulations can be seen in Figure 3.

First, an initial 3D Lagrangian position grid was constructed to map each particle back to its grid cell. Following Buisman et al. (2025), all substructures were excluded in this analysis (parent ID ≠ -1) and only host halos with a minimum virial mass of $M = 4 \cdot 10^{12} M_\odot/h$, corresponding to ∼ 800 particles, considered. The particles were then filtered to only those in host halos (assigned halo ID = internal halo ID). Then, these host halo particles were checked iteratively; if a particle belongs to a valid, high-mass host halo, its particle ID was mapped back to its original Lagrangian grid coordinate and its halo's axis ratios (b/a and c/a) values stored in the voxels.

Crucially, `ROCKSTAR` misses $\sim 2 \cdot 10^6$ particles of the $\sim 16 \cdot 10^6$ particles and does not assign them to halos. Inspecting their positions revealed that they are clearly clustered and therefore cannot be considered background particles and ignored. Consequently, `scipy.spatial.cKDTree` was used to find the five nearest host halos of the unidentified particles. If an unassigned particle is located within the virial radius of a nearby host halo, it was assigned to that host halo and the axis ratios stored in the particle's Lagrangian grid cell.

The resulting two Lagrangian label tensors were stored as arrays of shape (256, 256, 256) to further be used as labels for the CNN. Importantly, the individual axis ratios are given to the model as a two-channel target rather than the triaxiality parameter T directly. This prevents the network from learning degenerate solutions, as multiple distinct axis ratio pairs can map to the same T value.

2.2. Predicting halo axis ratios

The collapse and final geometry of bound structures is a highly non-linear process, shaped by anisotropic gravitational forces. While early theoretical and purely analytical works, such as the Press-Schechter formalism (Press & Schechter 1974) and its extension via excursion sets (Bond et al. 1991), focused on predicting the halo mass function based on spherical collapse thresholds

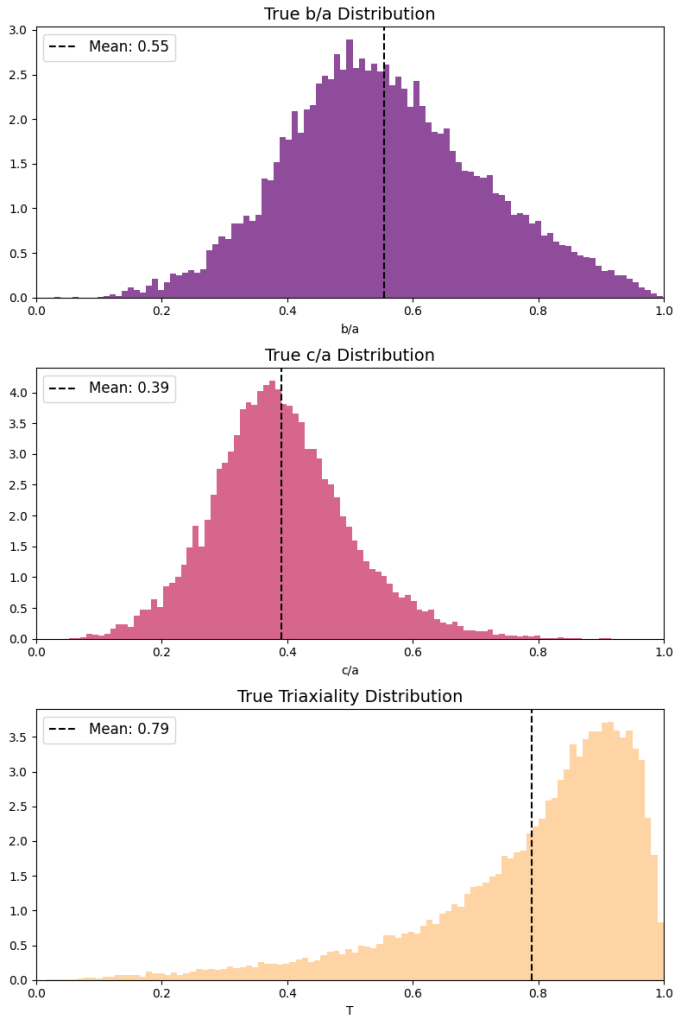


Fig. 3. Distribution of axis ratios and triaxiality for all selected host halos with $M \geq 4 \cdot 10^{12} M_{\odot}/h$ of all 110 simulations. The mean value of each distribution is indicated by a dashed black line.

($\delta \sim 1.686$) in a Gaussian random field, modern extensions also focus on modeling shape distortions.

Cosmic structures form via ellipsoidal collapse, where a dark matter halo's virialization condition depends on its initial ellipticity and prolateness (Sheth et al. 2001). Analytic frameworks derived from the primordial density peaks (Bardeen et al. 1986) have shown that the eigenvalues of the initial tidal shear field determine the non-spherical deformation of protohalos (Chan et al. 2012; Sheth et al. 2013). At the same time, high-resolution N-body simulations showed that material accretion occurs preferentially along cosmic filaments, which causes strong triaxiality onto the final virialized objects (Jing & Suto 2002).

Convolutional neural networks are a natural choice for predicting structure formation because the convolutional operator allows for predictions based on local features in the initial field. Stacking multiple such operations on different scales allows the model to learn the multiscale dependence of structure formation from the different features in the field. Moreover, CNNs capture non-local spatial gradients and directional anisotropic features. The architecture used in this work is a U-Net (Ronneberger et al. 2015) (also called a V-Net in 3D) and has been used successfully to make a variety of predictions in astrophysics (Buisman et al. 2025; Barthe-Gold et al. 2025; Wang et al. 2023). Compared to theoretical models, which select a handful of parameters considered important for the resulting structure, CNNs are able to make predictions based on all quantities encoded in the field. This makes them a powerful tool for predicting such structures, even if they come with the downside of having limited physical interpretability.

In the following sections, data processing and augmentation are introduced, followed by the introduction of the network hierarchy and finally, the loss function and training procedure.

2.2.1. Data processing

To accurately predict the halo geometry of each particle, enough training data is required for the model to learn what governs the complex processes that determine the final axis ratios. For this, the 110 N-body simulations, obtained as described in subsection 2.1, were randomly divided into 90 simulations reserved for training, 10 for validation, and 10 for testing. Their corresponding initial fields and final b/a and c/a axis ratios for each particle were then used as labels for the CNN.

As every simulation is a discrete realization of a fixed set of cosmological parameters, there is inherent variance in how much structure forms. To allow the learning process to understand this variance, the entire training set was normalized using a standard z-score.

$$x_{\text{norm}} = \frac{x - \mu_{\text{global}}}{\sigma_{\text{global}}} \tag{2}$$

μ_{global} was calculated by finding the mean density contrast value in the entire training simulation set. σ_{global} was calculated using

$$\sigma_{\text{global}} = \sqrt{\overline{\sigma^2} + \sigma^2_{\mu_{\text{global}}}} \tag{3}$$

This formula accounts for the difference in the variance ($\overline{\sigma^2}$) of the density field for the training set, as well as the difference that arises due to the variance in means ($\sigma^2_{\mu_{\text{global}}}$).

For the 90 training simulations, the global mean density is $-3.8 \cdot 10^{-10}$, while the standard deviation is 4.38.

To load, normalize, and prepare the training, validation and test data, classes called `CosmologyTrainDataset` and `CosmologyTestDataset` were implemented using `Pytorch` (Paszke et al. 2019). While both are similar in many functionalities, such as normalization based on the global training set properties, `CosmologyTrainDataset` additionally sub-samples the data cubes and includes data augmentation.

The entire training and evaluation process was run on the workstation Coxeter, which contains a single `NVIDIA GeForce RTX3090 GPU` with 24 GB of `VRAM`. A full volume box consists of 256^3 particles, which quickly exceed the computing capabilities, given the large CNN structure and the multiple simulations (8) processed per batch. Therefore, a function called `random_crop` was implemented, which selects a random point inside the box and returns a cropped cube of size 80^3. Additionally, this function needs periodic wrapping conditions, given the nature of such N-body simulations living on a torus. Furthermore, the `mmap` cache mode was used, which helps reduce the memory load.

As the entire label and simulation generation requires significant computational time and storage space, any possibilities to maximize the use of each simulation should be leveraged. The cosmological principle and isotropy allow data augmentation for each

simulation in two ways: random flipping along any of the three axes, as well as arbitrary rotations of the cube. This increases the initial simulation size of 90 to 4320 (a factor of 48).

These functionalities were implemented for the training set only to increase its size, while the test and validation set were evaluated with a smaller batch size (1). This allows the prediction of the full sized simulation volume, since the fully convolutional architecture places no constraint on input size.

2.2.2. CNN architecture

To achieve optimal results, the chosen network architecture must reflect the physical mechanisms driving the formation of structure. As mentioned in the introduction of this chapter, the formation and final geometry of a halo are determined by many factors that span large physical scales. Therefore, the machine learning architecture must reflect this multiscale influence in order to predict the final halo axis ratios of each particle.

Figure 4 showcases the 4-level U-Net architecture used in this work. The name stems from the nature of compressing and expanding the physical size of the cube and the so-called skip connections, resembling the shape of a U. In the encoder, the initial density field is compressed in physical size at each layer while the feature depth increases. The initially passed sub-volume contains 80^3 particles corresponding to a cube of side length 31.25Mpc/h. The network takes a single-channel input representing this initial dark matter density field. The convolution operators in the first layer output feature channels $n = 32$, doubling with each deeper layer as the features become increasingly abstract. The maximum compressed cube is in the lowest layer, called the bottleneck, after which the cube is up-sampled. The information is interpolated based on the higher compressed cube and the information from the skip connection. This repeats until the cube reaches the original size again, at which the voxel-per-voxel prediction is then made.

At each layer, the density field is convolved and an activation function is applied. The convolution was performed with `Conv3d`, with a filter of size $3 \times 3 \times 3$ and a stride of 1. The padding was chosen to be circular, motivated by the periodic wrapping of simulations. For the activation function, `LeakyReLu(0.05)` was used. This structure of convolution followed by an activation function is applied twice per layer. In the baseline architecture, a dropout layer `Dropout3d` is integrated into the encoder before the size of the cube is halved via `MaxPooling(2)`. The addition of a dropout operator should reduce the chance of overfitting (Hinton et al. 2012). However, through trial and error, the dropout rate was set to zero for the final network. As for the compression, `MaxPooling(2)` may not be the most cosmologically motivated compression choice. A more principled solution would apply a sharp cut-off in Fourier space (Flöss et al. 2024). An alternative would be a convolution operator with a stride of 2, but given the computational limitations and the scope of the project, the simple `MaxPooling` compression was chosen.

Maximum compression is achieved in the bottleneck, where the number of feature channels reaches $n = 256$ and the $3 \times 3 \times 3$ convolutional kernel spans a physical scale of 9.375 Mpc/h. Afterwards, the cube is inflated by a factor of two using `ConvTranspose3d` with a kernel of size $2 \times 2 \times 2$ and a stride of 2. Before convolving the field again, the information from the lower layer is concatenated with those provided by the skip connection at the same layer. Concatenating the two allows the subsequent convolution to draw on both the coarse and abstract features from the bottleneck and the fine spatial detail preserved by the skip connection. This repeats until the cube has the size of the original input, where, after convolution, a prediction for both axis ratios is made for each voxel via a convolutional operator with a kernel size of 1 and a stride of 1. This final layer maps the abstract features down to an output of two distinct channels, corresponding to the spatial distributions of the axis ratios b/a and c/a, respectively. Importantly, even though axis ratios are physically bounded between 0 and 1, the final layer uses a linear output rather than a restricted activation function like a `Sigmoid` or `HardSigmoid`. Through trial and error, it was found that adding a bounded activation function caused the network to stop learning very early in the training process. This is because these functions cause the gradients to vanish quickly near the boundaries, whereas a linear output allows smooth error gradients to flow backward through the network.

2.2.3. Training

The data set was divided into 90 independent simulations for training, 10 for evaluation and 10 for testing, each containing a 3D density field and its associated voxel-wise axis ratio labels. The model was optimized using Adam (Kingma & Ba 2017) with a learning rate of 10^{-4}. To enable neural network optimization, background voxels, originally represented as `NaN`, were mapped to a constant value of 0.0 during training.

The Mean Squared Error (MSE) was selected as the loss function, calculated voxel-by-voxel across the 3D grid and normalized by the total volume of the cube (80^3). The loss was computed independently for each axis ratio channel and the total loss was defined as the sum of both individual errors;

$$\mathcal{L}_{\text{total}} = \text{MSE}_{b/a} + \text{MSE}_{c/a}, \tag{4}$$

where the MSE for a single channel is defined as

$$\text{MSE} = \frac{1}{80^3} \sum_{j=1}^{80^3} (y_{\text{true},j} - y_{\text{pred},j})^2. \tag{5}$$

Training was performed on a single NVIDIA RTX 3090 GPU with 24 GB of VRAM, for a maximum of 1000 epochs, where one epoch corresponds to a full pass through all 90 simulations with random cropping and augmentation applied. An epoch took $\approx$ 14 seconds. An early stopping criterion was implemented, stopping the training process if no improvement in validation loss was observed for 40 consecutive epochs. This helps mitigate overfitting and was triggered after 520 epochs. Figure 5 shows the training and validation loss curves. Although there is a constant scaling factor present between the absolute values of the curves, their parallel trajectory confirms that the network is generalizing well and that there is no architectural overfitting. In total, the entire training process took a little more than 2 hours.

2.3. Evaluation diagnostics

The model's predictions are evaluated on multiple metrics. First, the following binary classification metric was used,

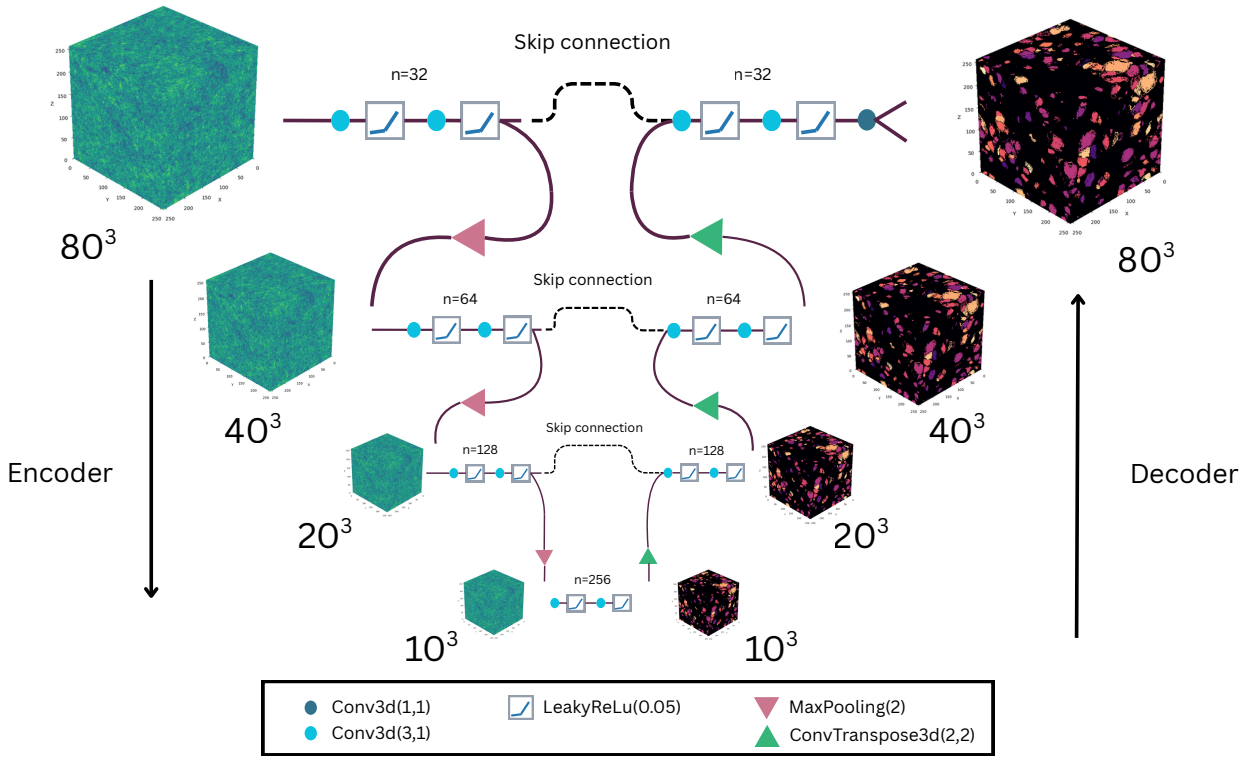


Fig. 4. 4-level U-Net structure. Starting from the top left, a random sub-volume of size 80^3 particles is selected from the training data. This cube is processed through convolutional operators, activation functions and pooling operators down to the next encoder layer. This process repeats, compressing the volume by a factor of two and increasing the channel depth n by a factor of two. At the bottleneck layer (lowest) the cube is maximum compressed and the channel depth maximized. Afterwards, the compressed cube's spatial dimension is increased by a factor of two using `ConvTranspose3d()`. The output of the lower level and the information of the level are combined and processed through convolutional and activation operators. This repeats until the cube has the starting size, where a final $1 \times 1 \times 1$ convolution maps the features to the two output channels.

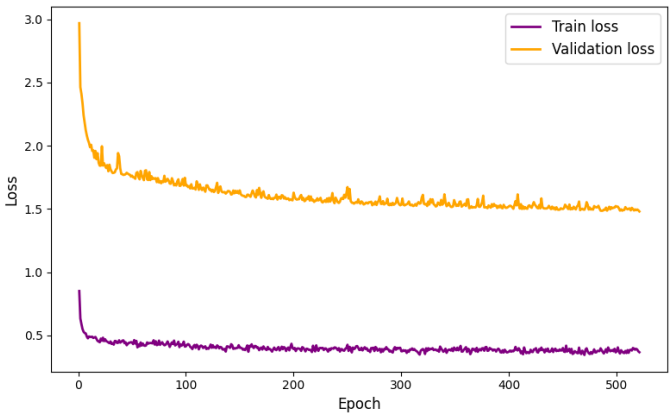


Fig. 5. Training (purple) and validation (orange) loss function for the training of the U-Net model.

$$\text{true positive rate} \quad \text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{true negative rate} \quad \text{TNR} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

$$\text{positive predictive value} \quad \text{PPV} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{accuracy} \quad \text{ACC} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FN} + \text{FP}}$$

$$F_1\text{-score} \quad F_1 = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}},$$

where a true positive (TP) is defined as a correctly identified collapsed particle and a true negative (TN) is a correctly classified background particle. Since the network utilizes a linear output layer, its final predictions are continuous rather than discrete probabilities. To turn these continuous values into a binary prediction during testing, a threshold of $\epsilon = 0.05$ is applied. Voxels with axis predictions below this threshold are classified as background, while voxels with values equal to or greater than ϵ are classified as collapsed.

Moreover, since the model directly predicts continuous halo axis ratios, the regression performance is evaluated using voxel-wise scatter plots of the axis ratios and triaxiality. In the most optimal case, this comparison leads to a diagonal line where $y_{\text{pred}} = y_{\text{true}}$.

3. Results

3.1. Predicting collapse and shape

First, the model was evaluated based on the binary classification of correctly identifying which particles collapsed and which did not. Table 1 shows the results of the binary classification metric using all 10 test simulations and a threshold of $\epsilon = 0.05$. The high TPR of 0.976 indicates that halo structures are overall successfully identified by the network. However, the lower PPV and F_1 values highlight a high rate of false positives, showing that the model tends to bleed shape predictions into the background.

To investigate the predictions visually, Figure 6 shows a slice of one test simulation. The first row shows the ground truth and the

Metric	Value
TPR	0.976
TNR	0.736
PPV	0.524
ACC	0.791
F_1	0.682

Table 1. Binary classification performance on the 10 simulation test set. A particle is considered background if its value is smaller than $\epsilon = 0.05$.

second row the predictions, with the halos colored by axis ratio value in the first two columns and by computed triaxiality in the last column. In the last row, the absolute difference between the truth and prediction are visualized. Notably, the model predicts the locations of the halos but introduces a gradient outside. The error plot highlights the stronger agreement to the true axis ratios in the center of the halos.

Figure 7 shows the predicted values as a function of ground truth for the axis ratios and computed triaxiality for 20 000 randomly sampled voxels. Clearly, the model fails to reproduce the optimal diagonal relation but predicts a broad range of values. The indicated coefficient of determination values R^2, computed across the full test simulation set, quantitatively confirm an anti-correlation and highlight the network's inability to capture the underlying geometric distribution.

To determine whether the model's poor regression performance is due to a lack of neural network capacity or a fundamental constraint, a controlled single-sample overfitting diagnostic was performed. For this, it was trained on a single simulation for 3000 steps with a learning rate of 10^{-4}. Figure 8 shows that the architecture is able to predict a range of axis ratios closer to the perfect fit line.

3.2. Predicting shape only

To isolate why the baseline network fails to reproduce halo axis ratios, the architecture was restricted to axis ratio predictions by masking out background voxels. The background voxels were assigned a value of `NaN` based on the ground truth and the loss function was modified to evaluate bound halo voxels only. Similar to the baseline model, it excludes both final activation functions and dropout layers. The model was trained as described in subsubsection 2.2.3 and stopped after 52 epochs (~25 minutes).

Analogous to the results in Figure 6, Figure 9 shows the resulting predictions for an example simulation slice. Despite isolating the foreground geometry, all halos are predicted to have identical, almost uniform values across the entire volume.

In Figure 10, the predicted axis ratios and computed triaxiality are plotted against their ground-truth values for 20 000 randomly sampled foreground voxels. The predictions collapse into narrow, horizontal bands and fail to track the ideal fit. The R^2 scores are close to zero across all three metrics and therefore quantitatively confirm an absence of correlation. This indicates that the network remains unable to extract genuine structural information, even when constrained to halo voxels exclusively.

4. Discussion

4.1. Choice of predicted halo property

This analysis was conducted as part of a larger project predicting halo properties using N-body simulations. While Jonas & Einramhof (2026) predicts collapse and halo mass across 12 mass bins, the parameter initially chosen for this work was the halo concentration parameter, given by the ratio of the virial radius and the scale radius, $c_{\rm vir} = R_{\rm vir}/r_s$ (Bullock et al. 2001). However, the halo concentration-mass relation (Correa et al. 2015) could not be recovered, indicating that the inner regions of the halos are not sufficiently resolved in the relatively low-resolution simulations used in this study. Consequently, it was necessary to reconsider the choice of predicted property.

Ultimately, halo shape was selected, as `ROCKSTAR` returns the axis ratios b/a and c/a for every halo. From these parameters, the triaxiality T can be computed, which is a single scalar description of halo shape, commonly used to analyze halo geometry (Giocoli et al. 2026). Moreover, the continuous nature of triaxiality naturally frames this as a regression problem, further differentiating this approach from the binned halo mass prediction in Jonas & Einramhof (2026) .

4.2. Data generation and CNN limitations

Both the dataset generation pipeline and the CNN architecture have limitations that impact the performance of this model.

Firstly, due to computational, memory and storage constraints, only 110 simulations were generated in total. This provides a significantly smaller training dataset than comparable works (e.g., Buisman et al. 2025). A direct consequence of this limited volume is the underrepresentation of high-mass halos, resulting in an imbalanced halo property distribution. Consequently, splitting the data into training, validation, and test sets requires care to ensure that each split contains a representative sample of the halos in the simulations.

Additionally, because `ROCKSTAR` fails to explicitly assign a substantial subset of dark matter particles to halos, a spatial assignment based on the virial radius was implemented. However, while this is a necessary requirement, geometric proximity alone does not guarantee gravitationally bound membership of the particles. Therefore, this potentially introduces marginal distortions into the ground-truth halo shape definitions. However, given the poor performance of this model, this boundary distortion is unlikely to be a significant issue.

Furthermore, computational limits forced the input domain to be split up into smaller 80^3 voxel sub-volumes. While this risks losing large-scale environment information, its impact on local shape regression should remain minimal provided the sub-volume fully encompasses the target halo's extent.

Regarding the architecture, operating without a final activation function leads to unphysical predictions, such as negative axis ratios. Conversely, introducing `HardSigmoid` as a bounding function causes severe vanishing gradients during training, which causes the network to be unable to optimize.

For the loss function, the model uses a standard Mean Squared Error (MSE) summed across the b/a and c/a channels. This is considerably simpler than the Earth Mover's Distance (EMD) loss used in Jonas & Einramhof (2026). Performance could likely be improved by adopting an adaptive, physics-informed

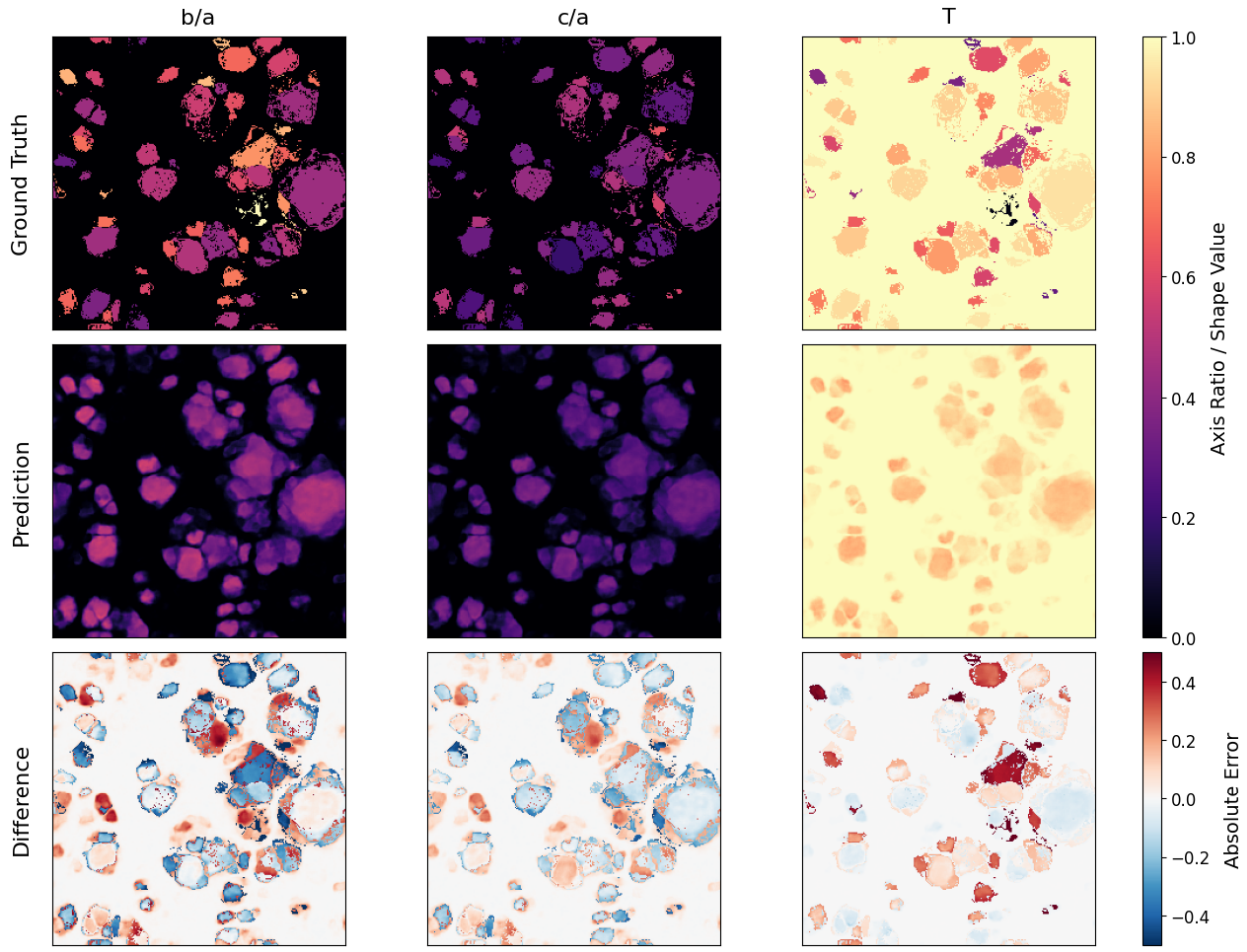


Fig. 6. Predictions for a slice of one test simulation. The first row shows the ground truth, the second row the predictions and the last row the absolute error between the two. The columns show the results for the axis ratios b/a, c/a and triaxiality T, respectively.

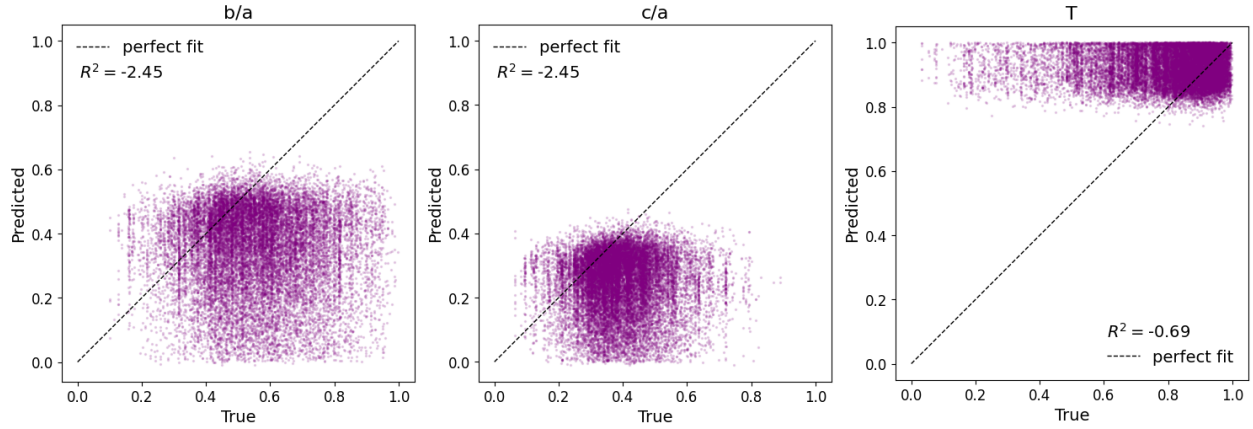


Fig. 7. Predicted values as a function of ground truth for the axis ratios b/a (left), c/a (center) and computed triaxiality (right) for a random sub-sample of 20 000 points. The dashed black line indicates the perfect fit. Every panel shows the computed R^2 value for all test simulations.

loss function that explicitly reweights background voxels, penalizes shape boundary errors or accounts for shape anisotropy.

Finally, the baseline 3D CNN is analogous to Jonas & Einramhof (2026), where it successfully performed discrete mass classification, but it may generally be ill-suited for continuous geometric regression. Therefore, alternative architectures, such as graph neural networks (GNN) (Delgado et al. 2026), may be a better fit for this task.

4.3. Predicting collapse and shape

The model was first evaluated on its ability to simultaneously identify collapsed regions and infer their geometric shape values from the initial density field.

To quantify collapse prediction performance, a binary classification scheme was applied and the resulting metrics are compared against literature values in Table 2. The model achieves a

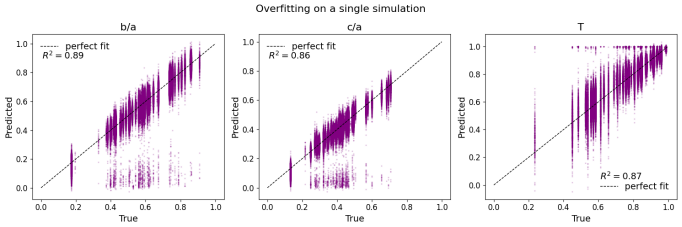


Fig. 8. Predicted values as a function of ground truth for the axis ratios b/a (left), c/a (center) and computed triaxiality (right) for a single simulation the model was trained on. The dashed black line indicates the perfect fit. Every panel shows the computed R^2 value for all test simulations.

TPR of ~ 0.976, notably higher than literature models. This indicates an unconservative prediction, where almost all true halo particles are successfully captured within predicted collapsed structures. Conversely, the TNR is substantially lower (~ 0.736) than literature values, demonstrating that a significant fraction of uncollapsed background particles are falsely assigned to halos. For this evaluation, a particle was classified as collapsed if its predicted axis ratio exceeded a threshold of 0.05, accounting for baseline prediction variance around true zero. In contrast, Buisman et al. (2025) define collapse probabilistically by summing predicted probabilities across mass classes above a 50% threshold. This fundamental difference directly influences this model's TPR and TNR. This over-prediction of collapsed volume severely degrades the PPV (~ 0.524), indicating that nearly half of all particles considered collapsed are false positives. Consequently, the F_1 score (~ 0.682) is lower relative to literature models. Finally, because cosmic volumes are dominated by background voxels, the reduced TNR directly lowers the overall accuracy.

Metric	López-Cano	Buisman	**This work**
TPR	0.838	0.732	0.976
TNR	0.883	0.956	0.736
PPV	0.838	0.822	0.524
ACC	0.864	0.907	0.791
F_1	0.838	0.774	0.682

Table 2. Binary classification performance on the 10 simulation test set, compared against literature models (López-Cano et al. (2024),Buisman et al. (2025))

Figure 6 visually inspects the prediction results for a slice of a test simulation. The model successfully captures large-scale collapsed regions, however, the predicted values deviate significantly from the ground truth. The error maps in the bottom row show that the predicted axis ratios align more closely with true values near halo centers, whereas errors increase significantly toward halo edges. This behavior is further demonstrated by the scatter plots in Figure 7, which show predictions clustering around or below the dataset mean (Figure 3). The distinct vertical bands demonstrate that for a given true label, the model outputs a range of underpredicted values. This is consistent with the gradients observed toward halo boundaries in Figure 6. Quantitatively, the poor fit is reflected in negative R^2 scores across all target metrics ($R^2 = -2.45$ for both axis ratios, and $R^2 = -0.69$ for triaxiality).

Finally, the single-sample overfitting experiment in Figure 8 confirms that the network architecture possesses sufficient capacity to learn better agreement with the ground truth. The predictions follow the perfect fit line, resulting in R^2 scores of ~ 0.9 for the axis ratios and triaxiality. Nevertheless, the vertical scattering is still prominent, indicating that even when memorizing a single volume, the model struggles to consistently resolve halo outer boundaries. These results demonstrate that the primary limitation is not the architectural capacity alone but rather a failure to generalize learned structural features across unseen simulations.

4.4. Predicting shape only

Figure 9 and Figure 10 show the results for the modified model evaluated exclusively on masked background voxels. Even with background noise isolated, the model remains unable to predict varying axis ratios. Instead, the collapse toward the mean values becomes much more pronounced. In Figure 10, the scatter is a tight, horizontal band centered at the mean across the entire domain of true values. The corresponding R^2 scores are close to zero, which quantitatively confirms absence of correlation. This mean-collapse behavior can be clearly seen in Figure 9, where halos display an almost uniform color from the center to the edges, indicating that the gradients have vanished completely.

5. Conclusion and outlook

In this work, a 3D U-Net-based CNN was used to predict dark matter halo geometry from initial linear density fields. In the baseline setup, where the model simultaneously predicted collapse and halo axis ratios, the network successfully identified the spatial structure of collapsed regions on large scales. However, it failed to accurately predict axis ratios. Instead, it predicted values near the dataset mean at the halo centers and artificial decreasing gradients toward the halo boundaries. A single-sample overfitting diagnostic demonstrated that the network has sufficient capacity to fit single volumes but struggles to generalize structural features across independent simulations. To isolate geometry prediction from background collapse, a second experiment masked out background voxels using ground-truth masks. This caused the predicted values to collapse even more tightly into a narrow distribution around the mean, resulting in $R^2 \approx 0$. These findings indicate that the current CNN architecture cannot extract physically meaningful halo axis ratios from initial density fields in this setup.

Nonetheless, predicting halo triaxiality from simulations is an active line of research. Alternative approaches, such as semi-analytic random walk models (Menker & Benson 2022) or hybrid networks applied to massive clusters (Delgado et al. 2026) have demonstrated that machine learning can successfully model halo geometry under appropriate formulations.

Further work could bin shapes into prolate, oblate and triaxial classes and investigate if the performance of the model changes in this simpler set-up. However, given the mean-collapse behavior observed in this architecture, such a transition would likely not be enough without also updating the network. Moreover, replacing the standard MSE loss with an adaptive or physics-informed loss function might prevent the network from settling into the global mean solution.

Despite the limitations in this work, CNNs paired with hydrodynamical simulations remain crucial for cosmology by decreasing computational costs and potentially breaking parameter degeneracies in future galaxy surveys. Motivated by recent advances in multi-zoom techniques and subgrid modeling (Maion et al. 2026), future architectures could use full hydrodynamical simu-

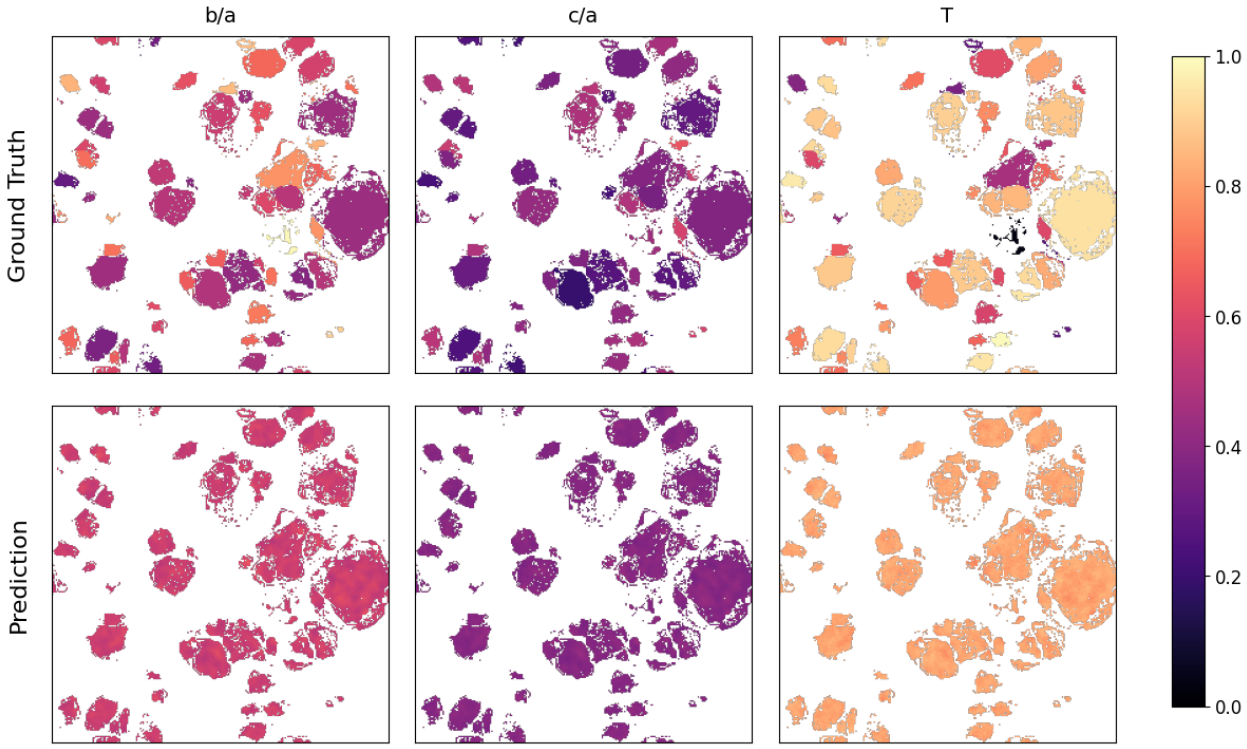


Fig. 9. Predictions of the network with masked background particles for a slice of one test simulation. The first row shows the ground truth and the second row the predictions. The columns show the results for the axis ratios b/a, c/a and triaxiality T, respectively.

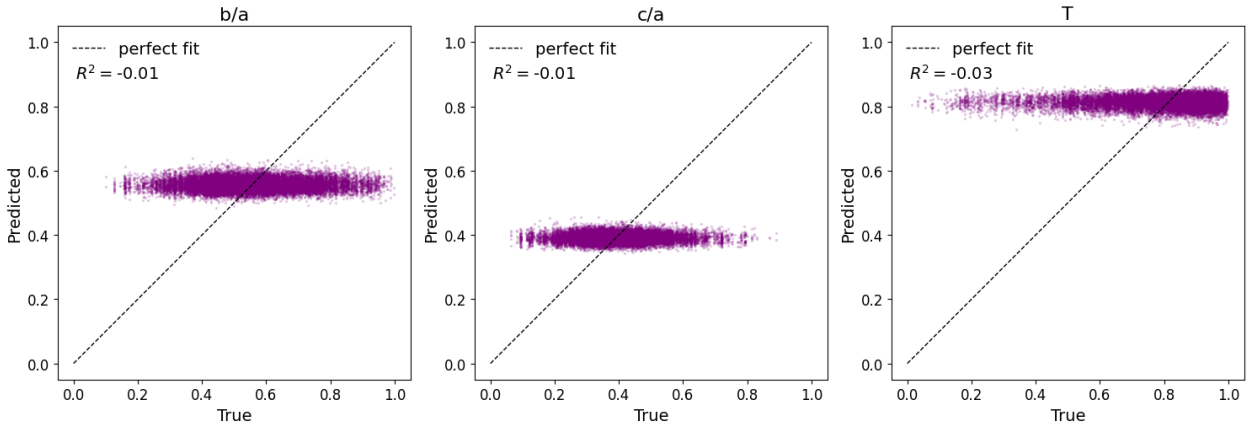


Fig. 10. Predicted values of the network with masked background particles as a function of ground truth for the axis ratios b/a (left), c/a (center) and computed triaxiality (right) for a random sub-sample of 20 000 points. The dashed black line indicates the perfect fit. Every panel shows the computed R^2 value for all test simulations.

lations to simultaneously separate subgrid physics, dark matter and cosmological parameters.

References

Bardeen, J. M., Bond, J. R., Kaiser, N., & Szalay, A. S. 1986, ApJ, 304, 15
Barthe-Gold, B., Nguyen, N.-M., & Thiele, L. 2025, Reconstructing the local density field with combined convolutional and point cloud architecture
Behroozi, P. S., Wechsler, R. H., & Wu, H.-Y. 2012, The Astrophysical Journal, 762, 109
Bonamigo, M., Despali, G., Limousin, M., et al. 2015, MNRAS, 449, 3171
Bond, J. R., Cole, S., Efstathiou, G., & Kaiser, N. 1991, ApJ, 379, 440
Buisman, J., List, F., & Hahn, O. 2025, Differentiable Halo Mass Prediction and the Cosmology-Dependence of Halo Mass Functions
Bullock, J. S., Kolatt, T. S., Sigad, Y., et al. 2001, Monthly Notices of the Royal Astronomical Society, 321, 559–575
Chan, K. C., Scoccimarro, R., & Sheth, R. K. 2012, Physical Review D, 85
Chen, H., Wang, J., Mao, T., et al. 2024, MNRAS, 532, 3947
Corless, V. L. & King, L. J. 2007, Monthly Notices of the Royal Astronomical Society, 380, 149
Correa, C. A., Wyithe, J. S. B., Schaye, J., & Duffy, A. R. 2015, Monthly Notices of the Royal Astronomical Society, 452, 1217–1232
Delgado, A. M., Ntampaka, M., Bose, S., et al. 2026, Estimating the triaxiality of massive clusters from 2D observables in MillenniumTNG with machine learning
Flöss, T., Coulton, W. R., Duivenvoorden, A. J., Villaescusa-Navarro, F., & Wandelt, B. D. 2024, Denoising Diffusion Delensing Delight: Reconstructing the Non-Gaussian CMB Lensing Potential with Diffusion Models
Giocoli, C., Despali, G., Moscardini, L., et al. 2026, Astronomy amp; Astrophysics, 706, A340
Hahn, O., List, F., & Porqueres, N. 2024, Journal of Cosmology and Astroparticle Physics, 2024, 063
Hinton, G. E., Srivastava, N., Krizhevsky, A., Sutskever, I., & Salakhutdinov, R. R. 2012, Improving neural networks by preventing co-adaptation of feature detectors

Jamieson, D., Li, Y., de Oliveira, R. A., et al. 2023, The Astrophysical Journal, 952, 145
Jing, Y. P. & Suto, Y. 2002, The Astrophysical Journal, 574, 538–553
Jonas, R. & Einramhof, L. 2026, Predicting structure formation and halo mass in cosmological simulations using a convolutional neural network
Kingma, D. P. & Ba, J. 2017, Adam: A Method for Stochastic Optimization
Lahiry, A., Bayer, A. E., & Villaescusa-Navarro, F. 2025, The Astrophysical Journal, 994, 129
Lee, C. T., Primack, J. R., Behroozi, P., et al. 2017, MNRAS, 466, 3834
List, F. & Hahn, O. 2024, Journal of Computational Physics, 513, 113201
List, F., Hahn, O., Flöss, T., & Winkler, L. 2026, Journal of Cosmology and Astroparticle Physics, 2026, 033
Lucie-Smith, L., Barreira, A., & Schmidt, F. 2023, Monthly Notices of the Royal Astronomical Society, 524, 1746–1756
Lucie-Smith, L., Peiris, H. V., Pontzen, A., Nord, B., & Thiyagalingam, J. 2024, Physical Review D, 109
López-Cano, D., Stücker, J., Ibañez, M. P., Angulo, R. E., & Franco-Barranco, D. 2024, Characterizing Structure Formation through Instance Segmentation
Macciò, A. V., Dutton, A. A., van den Bosch, F. C., et al. 2007, MNRAS, 378, 55
Maion, F., Angulo, R. E., Springel, V., Genel, S., & Bryan, G. L. 2026, Evaluating the flexibility of the MillenniumTNG galaxy formation model with multi-zoom re-simulations
Menker, P. & Benson, A. 2022, Monthly Notices of the Royal Astronomical Society, 516, 4383–4396
Paszke, A., Gross, S., Massa, F., et al. 2019, in Advances in Neural Information Processing Systems 32, ed. H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, & R. Garnett (Curran Associates, Inc.), 8024–8035
Press, W. H. & Schechter, P. 1974, Astrophys. J., v. 187, no. 3, pp. 425-438
Rampf, C., List, F., & Hahn, O. 2025, Journal of Cosmology and Astroparticle Physics, 2025, 020
Ronneberger, O., Fischer, P., & Brox, T. 2015, U-Net: Convolutional Networks for Biomedical Image Segmentation
Sheth, R. K., Chan, K. C., & Scoccimarro, R. 2013, Physical Review D, 87
Sheth, R. K., Mo, H. J., & Tormen, G. 2001, Monthly Notices of the Royal Astronomical Society, 323, 1–12
van Daalen, M. P., Angulo, R. E., & White, S. D. M. 2012, Monthly Notices of the Royal Astronomical Society, 424, 2954
Wang, Z., Shi, F., Yang, X., et al. 2023, Science China Physics, Mechanics amp; Astronomy, 67
Zentner, A. R., Kravtsov, A. V., Gnedin, O. Y., & Klypin, A. A. 2005, The Astrophysical Journal, 629, 219–232